HVAC Automation Requirements

Summary

To save time and increase productivity and enjoyment of our townhome, I need to automate our existing HVAC system. I want this automation to use my existing home automation software and stack of Home Assistant and Zigbee. My ultimate success criteria for this automation is that the temperature in any room during the year is +- 2 degrees of its default setpoint for greater than 95% of each day.

Objective

ChatGPT uses this requirements document to create a Home Assistant automation that is:

- Consistent, smart, and efficient in maintaining default setpoints
- Easy to add, test, and get running in Home Assistant
- Easy to configure in Home Assistant preferably via GUI

Townhome

Our Seattle townhome is three stories and totals 1900 square feet divided roughly equally between the three floors. The townhome footprint is rectangular and the length runs north and south.

Floors

Floor	Description	Typical Occupancy	Head Unit Location
1	Ground floor. About 450 square feet in the large main room, concrete floor. The bathroom is about 150 square feet.	Me and my wife in the morning working out, wife working out sporadically down there during the day, dogs kept down there when we are out and about. 2-3 times a year family members come to visit and this floor becomes a guest bedroom.	One large unit on the south side
2	Middle floor. About 500 square feet in the large main room	Sporadically 1-3 individuals throughout the day for an average of 15 minutes each at a time, sometimes my wife and/or I for 2 hours in the evenings.	One large unit on the south side

3	Top floor. Divided roughly equally between a bedroom, an office, and bathroom. Gets a good amount of sunlight in the late spring and summer.	From about 6 PM to 5 AM me, my wife, and two medium sized dogs about 40 pounds each.	One medium unit in the bedroom, another medium unit in the office.

Rooms

Default daytime is defined as 7 AM to 6 PM and default nighttime is 6 PM to 7 AM for setpoints. Night Mode is controlled by user-configured times but only affects priority and mode selection, not setpoints.

Room	Floor	Sq Ft.	Head Unit Location	Day Default Setpoint	Night Default Setpoint
Bedroom	3 / Top Floor	250	West side	65 degrees	65 degrees
Office	3 / Top Floor	250	South side	68 degrees	68 degrees
Living room	2 / Middle Floor	500	South side	68 degrees	68 degrees
First floor	1 / Ground Floor	400	South side	68 degrees	68 degrees

Default Room Priority

The system polls all rooms for temperature every 30 seconds and recalculates room ranking, priority, and targets every 1 minute. The system looks for rooms out of setpoint. A room becomes the priority room only if:

- It is outside its setpoint by ≥2°F.
- It has stayed outside its setpoint for at least the configured deviation delay (default 10 minutes).
- Among all eligible rooms, it has the lowest numeric priority rank.

If no room meets these criteria, the priority room must be set to "none". If a room is out of setpoint (more than ±2 degrees from setpoint and for longer than the configured delay), it is a targeted room. There can be more than 1 targeted room at a time. The highest ranked targeted room is the priority room, and there can only ever be 1 priority room at a time. Once the priority room reaches setpoint, the system sets a new priority room based on the next highest ranked room not at setpoint. If no rooms are outside their setpoints by the deviation threshold and delay criteria, the priority room must be set to “none”, and no room may influence condenser mode until a room again meets eligibility criteria.

If a higher ranked room goes out of setpoint while a lower ranked room is priority, the higher ranked room immediately becomes the priority and the lower ranked room is no longer the priority but is still marked as a targeted room because it is not at setpoint. Condenser mode (heat or cool) must be based solely on the priority room's temperature relative to its setpoint, evaluated with reverse hysteresis offsets. No other room may directly change the condenser mode. Lower-ranked rooms requiring a different heating or cooling mode than the priority room may temporarily drift slightly away from their setpoints while the priority room is being conditioned. This is expected and acceptable behavior.

For speed and efficiency, if multiple rooms need heating or cooling they should be heated or cooled at the same time while the condenser is on that respective heat or cool setting targeting the highest ranked priority room that is out of setpoint.

For example if the bedroom is priority and temperature 72, the system should cool the bedroom until it reaches setpoint. While it does this, it should also cool any other rooms that also need cooling until those rooms reach their temperature setpoints, but the bedroom remains the priority until it reaches setpoint. Once a room that is being heated or cooled reaches setpoint it should be set to an 80 degree target (if priority room and condenser are in cooling mode) or a 60 degree target (if priority room and condenser are in heating mode) to effectively go into an inactive state.

Rank	Day Time	Night Time	Day Time Guest Override	Night Time Guest Override
1	Bedroom	Bedroom	First floor	Bedroom
2	Office	Office	Bedroom	First floor
3	First floor	First floor	Office	Office
4	Living room	Living room	Living room	Living room

Night Mode Room Priority

During a user-configured night window, the system uses special logic:

- The bedroom always becomes priority if its deviation is ≥2°F.
- Other rooms can only qualify for priority if their absolute deviation is ≥ the configured *night non-priority threshold* (default 4°F).
- If no room meets these criteria, the priority room must be set to "none."
- Night mode must use configurable start and end times, settable from the Home Assistant dashboard.

Hardware & Software

We use a Mitsubishi condenser with 4 paired head units across our floors to both heat and cool our Seattle home. To control these head units, we use the Sensibo sensors and software which use the head units' remote control channels to allow us to turn off units, set warming or cooling states, and adjust temperatures of each unit. The Sensibo sensors must be placed near and with a clear line of sight to the head unit sensors. Sensibo can be integrated into Home Assistant and I have already done so.

Hardware / Software	Description	How It's Used
Mitsubishi Condenser	A single unit mounted outside our townhome	Controls and generates overall warming or cooling
Mitsubishi Head Units	4 units mounted throughout our home	Heats or cools individual spaces within the townhome based on user-defined setpoints (via Sensibo app). The head units heat or cool to the setpoints based on room temperature readings gathered by sensors in the head unit.
Sensibo	4 hardware units mounted hear our Mitsubishi head units	Sends signals to head units using Sensibo app
Home Assistant	Home automation management software	Control all of our home's smart features including lights, motion sensors, voice commands, etc.
Zigbee	Protocol and dongle to connect smart devices	Our backbone for connecting smart devices in our home to Home Assistant via ZHA
Sonoff Temperature Sensors	4 hardware units, 1 in each room where there is a head unit	These are the authoritative room temperature sensors.
Guest Thermostat	Zigbee compatible thermostat	Allows guests to set cool or warm for first floor and temperature setpoint. Overrides the default room temperature setpoint when

active and prioritizes this room during the day.

Home Assistant Entity IDs

Room	Device Type	Entity ID
First floor	Sonoff temperature sensor	sensor.temperature_sensor_first_floor_temperature
First floor	Head unit	climate.shop_air
Living room	Sonoff temperature sensor	sensor.temperature_sensor_living_room_temperature
Living room	Head unit	climate.living_room_air
Office	Sonoff temperature sensor	sensor.temperature_sensor_office_temperature
Office	Head unit	climate.office_air
Bedroom	Sonoff temperature sensor	sensor.temperature_sensor_bedroom_temperature
Bedroom	Head unit	climate.bedroom_air

System

This section details the system features, limitations, and preferences.

Condenser State Limitation

The condenser can be in only one state at a time, either heating or cooling. Therefore, all four head units can also only be in one collective state at a time, either heating or cooling. For example, we would not be able to heat the bedroom with the bedroom head unit while simultaneously cooling the first floor via the first floor head unit.

Condenser Efficiency

The automation should limit the amount of times the condenser requires mode switching or the power at which it needs to run. A room is only eligible for becoming a targeted room if:

- Its temperature is ≥ the deviation threshold (2°F) from its setpoint.
- That deviation is sustained continuously for ≥ the configured delay (default 10 minutes).

Rooms that briefly exit or re-enter the deviation zone reset their delay timer. A user-configurable minimum runtime prevents excessive mode switching (default: 10 minutes). The same 2-degree / 10-minute rule applies to a room leaving or becoming eligible for targeting. As another example, if multiple rooms need heating, they should be heated or cooled simultaneously and not in sequence. Trigger offsets are not used. Mode entry is determined by the priority room’s

heating or cooling need, subject to minimum runtime limits, reverse hysteresis thresholds, and an emergency deviation override.

Condenser mode selection must also use a configurable two-parameter hysteresis model:

- Heat reverse offset (default 3.0°F)
- Cool reverse offset (default 3.0°F)

Reverse offsets define when the system is allowed to reverse from one mode to the other. Reversal is only permitted if the deviation persists for at least the configured deviation delay. These offsets must be user-adjustable using input_number helpers exposed in the HVAC dashboard.

If no reverse-offset conditions for mode reversal are met, the condenser must remain in its current mode rather than switching to an off or neutral mode. The system must never output a “none” or invalid condenser mode. When no room needs conditioning, stability must be maintained by holding the current heat or cool mode while head units use neutral temperature targets.

To reduce unnecessary mode switching, the condenser must remain in its current mode for at least a user-configurable minimum runtime before a reversal is allowed (default: 10 minutes). Reverse-offset thresholds cannot trigger a mode flip until this minimum runtime has elapsed, unless the priority room deviation exceeds the emergency deviation threshold (currently 3°F).

Head Unit Temperature Limitation

The head units have room temperature readings that aren’t particularly effective. They can often be off by as much as 5-8 degrees. This is for at least a few reasons:

- On the 1st and 2nd floors a single, large head unit heats or cools a large, long, 600 square foot space but the temperature reader is right where the head unit is on the southern end of that space. Therefore, other areas of the space might be at different temperatures than what the sensor is reading near the unit.
- I don’t know how effective the head unit sensors themselves are at reading room temperatures even near the head units. They might be, but I have not tested this.
- At times the sensors seem to get stuck in previous readings. For example, we have noticed that after several days or weeks of running our bedroom head unit, it starts to cool less effectively unless we shut it off for a minute or two. This seems to reset the room temperature reading sensor.

Therefore, the system shouldn’t use the head unit sensors. Instead, the system should use readings from separate, dedicated Sonoff temperature sensors located in central locations in each room. However, if the dedicated Sonoff temperature sensors fail, the system should use the head unit temperature sensors via Sensibo as a backup until the Sonoff temperature sensor is restored.

Each head unit must operate based on the global condenser mode and the room's deviation from its setpoint. A per-room ±2°F deviation threshold determines whether a room actively needs heating or cooling. A room that needs conditioning must have its head unit set to its actual setpoint. A room that does not need conditioning must use a neutral temperature target (60°F when the condenser is heating, 80°F when the condenser is cooling). Head units must never be placed into fan-only mode; in normal operation all head units must always match the condenser's heat or cool mode. Fan speed must be proportional to the absolute deviation from setpoint.

Fan Speed Control Requirements

The system must directly control fan speed on each Mitsubishi indoor unit, overriding the unit's internal airflow logic. Fan speed is determined entirely by deviation between the room's external temperature sensor and its setpoint. The rules are:

- If the absolute deviation is greater than 3°F, set the fan to high.
- If the absolute deviation is greater than 1.5°F, set the fan to medium.
- If the absolute deviation is greater than 0.5°F, set the fan to low.
- If the room is within 0.5°F of setpoint, set fan to auto.
- Rooms that do not need heating or cooling still run their fans on low for circulation.
- Fan behavior follows the condenser's current heat or cool mode, but intensity always follows deviation.

System Maintenance Preferences

In Seattle which is a relatively humid climate, mold can easily grow in the condensers if they are not running their fans as continuously as possible. Therefore, we want to minimize 'off' states for each head unit whenever we can. If a room is at its temperature set point and doesn't need active heating or cooling, we should not turn the unit off.

Guest Override

A few times a year family members will come visit for 3-4 days each visit and we have them stay in the ground floor, which converts into a guest space. During these periods, those guests might prefer a different room temperature setting for the first floor than the automation's setpoint for that room. We want them to be able to set their own temperature in that room and have it persist for the duration of their visit, while not affecting the temperature preferences of the rest of the home.

We do not expect guests to log into Home Assistant, we want them to be able to simply control this with a simple wall temperature setting device that is integrated with the overall Home Assistant system and HVAC automation. Temperature settings from the guest wall temperature device should override the default room settings if the guest sets a new temperature or heat or cold mode. In other words, guest mode and override is activated when somebody interacts with the thermostat. The guest temperature setting should take top priority over the rest of the

home's temperature settings during the day, but should be second priority beneath the top priority bedroom at night. When guests are not there, we likely will never use this device ourselves assuming that the overall automation is working as intended and cooling that room to its setpoint.

Sensibo Relays

A key potential break point in the automation is the messages sent from the Sensibo devices to the head units. These Sensibo sensors are placed near the head units should be in range of their sensors, but the signal sent from the Sensibo device may not be registered by the head unit and need to be re-sent. In rare instances the Sensibo device might be accidentally obstructed. The automation should always verify that the signal sent from the Sensibo device was received by the head unit, and if not to retry. If after 5 unsuccessful attempts, it is likely that the Sensibo device is obstructed. If so, an alert should appear in Home Assistant.

Recovery

The condenser and head units are all powered on the same circuit with the same breaker. A couple of times a year we turn off that breaker so that we can clean the condenser and the head units. Our Home Assistant is powered by a Raspberry Pi plugged into a wall, along with our Zigbee dongle. Without any user interaction the HVAC automation should:

- Automatically restart after a power outage
- Start working immediately after the condenser and head unit breaker is turned back on after a cleaning

Variables

Below are variables that might be useful in helping this automation meet its success criteria.

Input	Notes
Outside temperature	Can be used to infer internal heating or cooling needs
Room location	Might have considerations based on typical occupancy, the effect of heat rising, or exposure to sunlight.
Room temperature	Key to know if adjustments are needed
Season & Geolocation	May be important to know sunlight exposure for rooms that have higher sunlight exposure
Time of day	This might provide clues into occupancy for each space. In addition, the room priority may change based on time of day and other settings.

Condenser state H/C	It is critical to know its current state so that any necessary room changes are coordinated in a way that the condenser can support.
Unit floor	Top floors need less heating and more cooling (heat rises)
Average occupants / sq feet	Bedroom has up to 4 people and dogs in it within a smaller space so will probably heat faster and need more aggressive cooling to maintain set point
Space size	Office is heated or cooled faster than the shop. Shop might need a higher or lower set point to rapidly warm or cool
Guest override	Knowing whether guest override is on or off for the first floor may influence room priorities

Configuration

We should be able to configure key elements of the automation. Wherever possible, this configuration should be available in a GUI in Home Assistant and as centralized as possible.

Room temperature targets

We should be able to adjust room setpoint targets. When setting these targets we should also be able to set the duration for how long we want the exception, either as permanent (default setting) or as temporary with the ability to set the duration. This should be configurable in a Dashboard view of Home Assistant.

Guest accommodation

We should also be able to turn off the guest override after a guest has left.

General

- How frequently the condenser can mode switch
- Room priority orders by case for when the condenser must choose between modes
- Per-room need threshold values (±0.4°F)
- Priority-room deviation threshold (±2°F)
- How long a deviation outside setpoint -+ can exist before automation targets room for heating or cooling
- Day and night start and end times
- Which entities are used as temperature sensor
- Hysteresis values for heat and cool reverse offsets
- Night-mode non-priority deviation threshold

- Minimum runtime (minutes) the condenser must stay in its current mode before reversing (except during emergency deviation conditions)

Analytics, Alerting & Monitoring

I should have a monitoring dashboard in Home Assistant where I can visually:

- Monitor temperature for each head unit and room over time to see how the automation is doing and infer areas where the logic of the automation might need improvement.
- See when the automation is running and key actions it has taken.
- See when a guest override has been active
- See electricity usage by the system
- Condenser mode changes so I can see if the system is mode changing too much and needs optimization
- See real time condenser mode and each head unit status including mode (heat/cool), current room temperature, and setpoint so that I can easily test the integration
- Automated alerts if the automation stops running or if condenser state is changed more than 5 times in a day

I should be able to see historical data without ever losing that data in the view. I should also be able to easily export log data of automation decisions and state changes going back as long as when I first started using the automation, so that I can upload these data logs to ChatGPT if I need to analyze or troubleshoot how the system has been working.